

SYSTEM DESIGN DOCUMENT

StockSense

System Workflow, Architecture & Use Case Diagrams

AI-Powered Inventory Shortage Predictor for Small Businesses

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Project Guide

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1. Introduction

This document presents the system workflow, architecture, and use case diagrams for StockSense. It illustrates how data flows through the system, how components interact, and how different users (actors) engage with the platform's features.

2. System Workflow

At a high level, StockSense follows this end-to-end workflow:

- Sales and stock data is collected from the business owner's point-of-sale records and manual stock entries.
- The data is cleaned, validated, and transformed through a preprocessing pipeline.
- Engineered features (sales velocity, seasonality, trends) are fed into a trained machine learning model.
- The model predicts the likelihood and expected timing of a shortage for each product.
- Predictions above a configured risk threshold trigger the Alert & Notification Engine.
- The backend exposes predictions, alerts, and historical trends via a REST API to the web dashboard.
- The business owner views the dashboard, receives alerts, and takes reorder action — closing the loop.

3. System Architecture

The architecture is organized into four layers — Data Sources, Processing & ML, Application, and Presentation — as shown below.

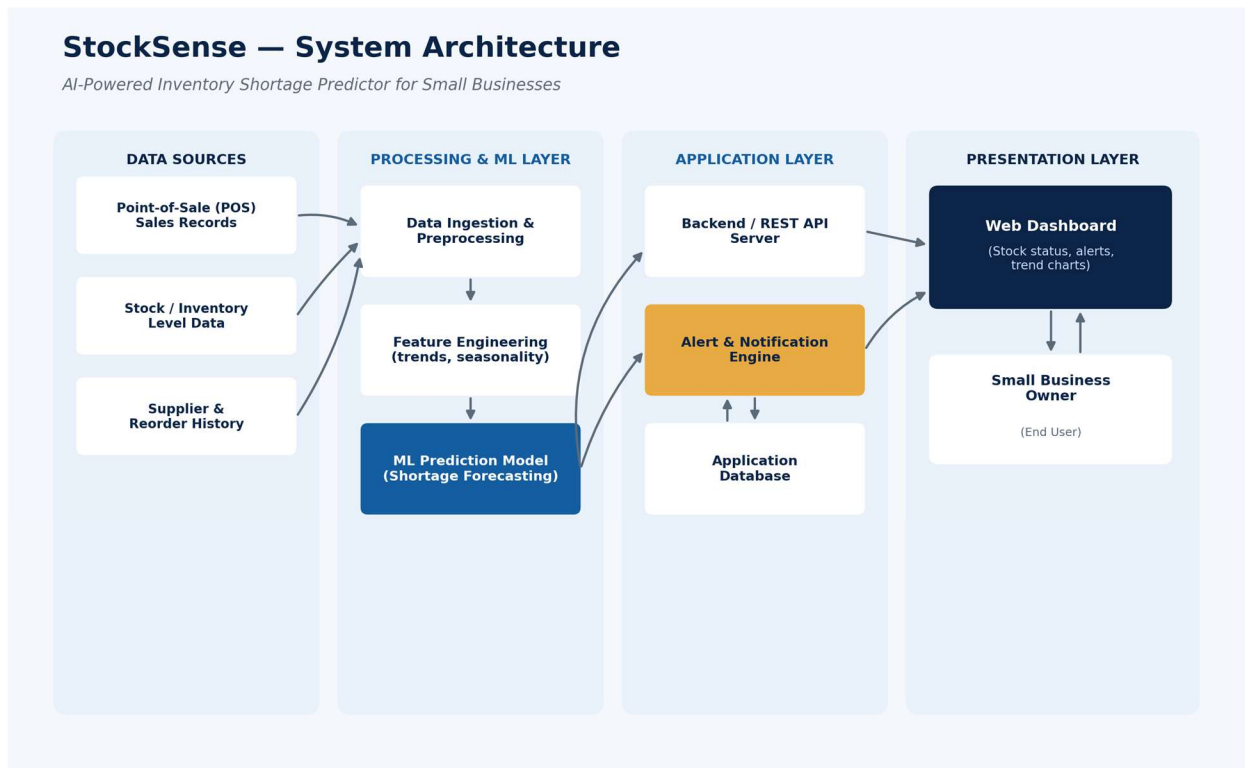


Figure 1: StockSense System Architecture Diagram

3.1 Layer Description

Layer	Responsibility
Data Sources	Captures POS sales records, stock/inventory levels, and supplier/reorder history.
Processing & ML Layer	Cleans and transforms raw data, engineers features, and runs the shortage prediction model.
Application Layer	Exposes REST APIs, manages the alert/notification engine, and stores data in the application database.
Presentation Layer	Web dashboard consumed by the small business owner to view stock status and alerts.

4. Use Case Diagram

The use case diagram below identifies the two primary actors — Business Owner and System Administrator — and the key interactions each has with the StockSense system.

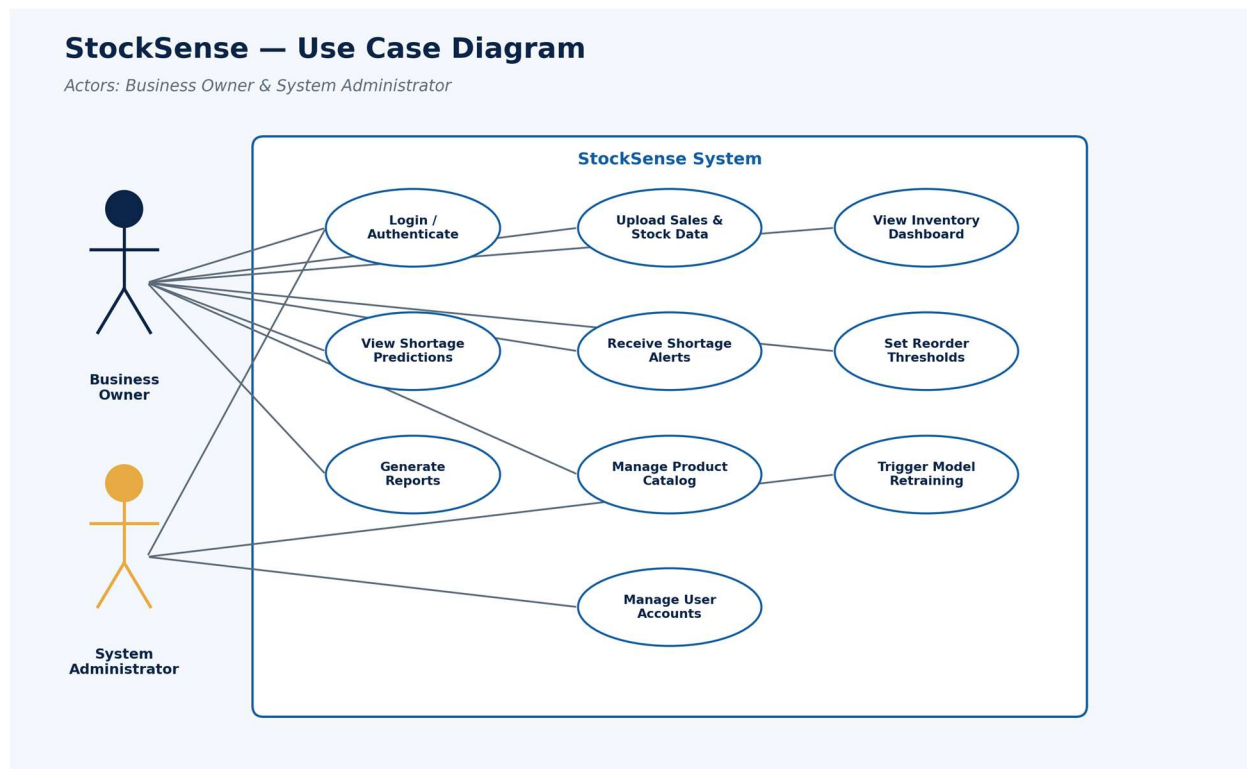


Figure 2: StockSense Use Case Diagram

4.1 Use Case Summary

Use Case	Actor(s)	Description
Login / Authenticate	Business Owner, Admin	Secure sign-in to access role-specific features.
Upload Sales & Stock Data	Business Owner	Add or sync sales transactions and stock levels.
View Inventory Dashboard	Business Owner	View current stock status and shortage risk.

Use Case	Actor(s)	Description
View Shortage Predictions	Business Owner	See per-product predicted shortage timing.
Receive Shortage Alerts	Business Owner	Get notified when a product nears a shortage threshold.
Set Reorder Thresholds	Business Owner	Configure custom stock levels that trigger alerts.
Generate Reports	Business Owner	Export basic inventory and prediction reports.
Manage Product Catalog	Business Owner	Add, edit, or remove products in the system.
Trigger Model Retraining	Admin	Initiate retraining of the ML model with new data.
Manage User Accounts	Admin	Create and manage system user accounts and roles.